STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 10B

Description

10 b) Write R program to find Non Linear Least Square with the sample data and visualize the regression graphically

Procedure and Overvation:

DESCRIPTION:

The goal of both linear and nonlinear regression is to adjust the values of the model parameteres to find the line (or)

curve that comes closest to your data and finding these values we will be able to estimate the response variable wit h good accuracy.

In least square regression to establish a regression model in which the sum of the squares of the vertical distances of different

points from the regression curve is minimized.

We generally start with a defined model and assume some values for the coefficients, we then apply the nls() of

R to get the more accurate values along with the confidence intervals.

SYNTAX:nls(formula,data,start)

where

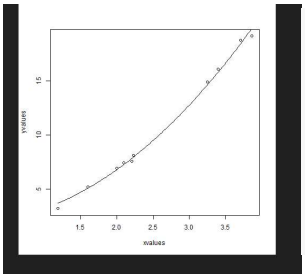
formula is a nonlinear model.It includes variab;es and parameters.

data is a data frame used to evaluate the variables in the formuls.

start is a named list or named numeric vector of starting estimates.

SOURCE CODE:

```
xvalues <- c(1.6,2.1,2,2.23,3.71,3.25,3.4,3.86,1.19,2.21)

yvalues <- c(5.19,7.43,6.94,8.11,18.75,14.88,16.06,19.12,3.21,7.58)

#Give the chart file a name.

png(file="nls.png")

#Plot these values.

plot(xvalues,yvalues)

#Take the assumed values and fit into the model.

model<-nls(yvalues~b1*xvalues^2+b2,start=list(b1=1,b2=3))

#Plot the chart with new data by fitting it to a prediction from 100 data points.

new.data <- data.frame(xvalues=seq(min(xvalues),max(xvalues),len=100))

lines(new.data$xvalues,predict(model,newdata=new.data))

#Save the file.

dev.off()

#Get the sum of the squared residuals.

print(sum(resid(model)^2))

#Get the confidence intervals on the chosen values of the coefficients.

print(confint(model))
```




```
> #Save the file.
> dev.off()
null device
          1
> source("C:/Users/student/Desktop/20R01A6753/week10b.r")
[1] 1.081935
Waiting for profiling to be done...
       2.5%    97.5%
b1 1.137708 1.253135
b2 1.497364 2.496484
> |
```

VIVA RESPONSES

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